

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Fourth Semester of B. Tech. (EC) Examination

May 2014

EC 208 Analog Electronic Circuits

Date: 07.05.2014, Wednesday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 70

Instructions:

1. Section I and II must be attempted in separate answer sheets.
2. Use of scientific calculator is allowed.

SECTION – I

- Q - 1 (a) Draw circuit diagram of Wien bridge oscillator. [03]
(b) Mention the features of power Amplifier. Explain class-A power amplifier with necessary illustrations. [04]
- Q - 2 (a) Describe meaning of a current controllable negative resistance (NR) device and voltage controlled negative resistance device. Give example of each types of NR devices. [02]
(b) Explain RC Phase Shift oscillator with necessary diagram. [06]
(c) Describe current sweep circuit and waveforms for inductor current and collector voltage. [06]

OR

- Q - 2 (a) Describe meaning of a negative resistance device? When does a device possess negative resistance? [02]
(b) Describe basic principles of Miller and Bootstrap time-base generators with neat block diagrams. [06]
(c) Describe linearity correction through adjustment of driving waveform for producing linear current ramp in an inductor. [06]
- Q - 3 (a) Define terms: (a) Time Base Generator (b) Sweep time (c) Restoration Time (d) Slope Error (e) Displacement Error (f) Transmission Error. In addition, mention equation relating slope error, displacement error and transmission error for linear time base generator. [07]
(b) Mention the advantage of tracking type Analog to Digital Converter over counter type Analog to Digital Converter. Using logic diagram describe tracking type Analog to Digital Converter. [07]

OR

- Q - 3 (a) Describe basic principles of Miller and Bootstrap time-base generators with neat block diagrams. [07]
(b) Describe successive approximation type ADC using neat diagram. Explicitly mention advantage of successive approximation type ADC. [07]

SECTION – II

- Q - 4 (a)** Determine regulated voltage in the circuit of Figure 1. Consider $R_1 = 240\Omega$, variable resistor $R_2 = 2.4\text{ k}\Omega$, $V_{ref} = 1.25\text{ V}$ and $I_{adj} = 100\text{ }\mu\text{A}$. [03]

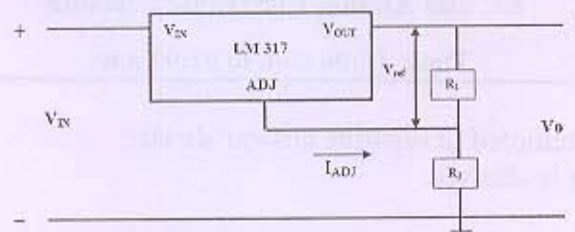


Figure 1

- (b)** Give full name of CMRR. Derive equation of CMRR. [04]
- Q - 5 (a)** Explain emitter-coupled differential amplifier. In addition, describe emitter couple differential amplifier having constant current source. [04]
- (b)** With the help of the $V - I$ characteristic and load line, explain the operation of tunnel diode bistable circuit. [05]
- (c)** Draw block diagram of feedback amplifier. Identify each block and state its function. [05]

OR

- Q - 5 (a)** Consider the situation where the first set of signals is $V_1 = +50\mu\text{V}$ and $V_2 = -50\mu\text{V}$ and the second set of the signals is $V_1 = 1050\mu\text{V}$ and $V_2 = 950\mu\text{V}$. If the CMRR is 100, calculate the percentage difference in output voltage obtained for the two sets of input signals. [04]
- (b)** With the help of the $V - I$ characteristic and load line in single diagram, explain the operation of (i) monostable (ii) bistable (iii) astable operation of a current controlled negative resistance devices. [05]
- (c)** Define desensitivity(D) with reference to feedback amplifier. Enlist four characteristics of an amplifier which are modified by negative feedback. [05]

- Q - 6 (a)** Give two practical applications of clipper circuits. Draw the output of Figure 2. Consider $V_{dc-1} = V_{dc-2} = 3\text{ V}$ and ideal diodes. [07]

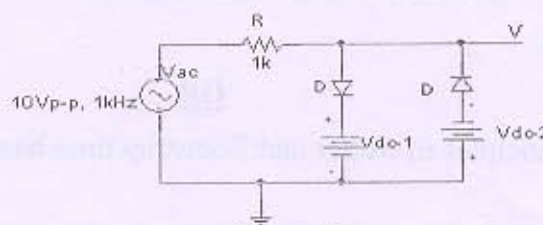


Figure 2

- (b) Mention two practical applications of clamper circuits. Draw the output waveform for the given circuit of Figure 3. Consider ideal diodes. [07]

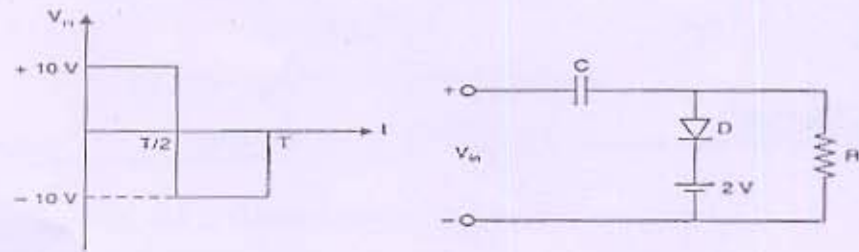


Figure 3

OR

- Q - 6 (a) Describe how transistor can work as a clipper. Draw the output waveform of Figure 4. Consider ideal diodes and $V_2=10V$ Peak to Peak. [07]

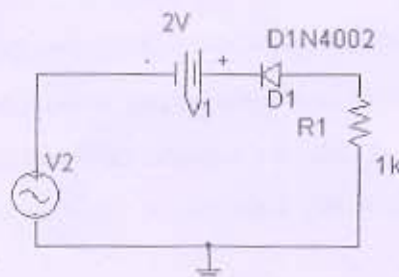


Figure 4

- (b) Mention two differences between clipper and clamper circuits. Draw the output waveform for the circuit of Figure 5. Consider ideal diodes. [07]

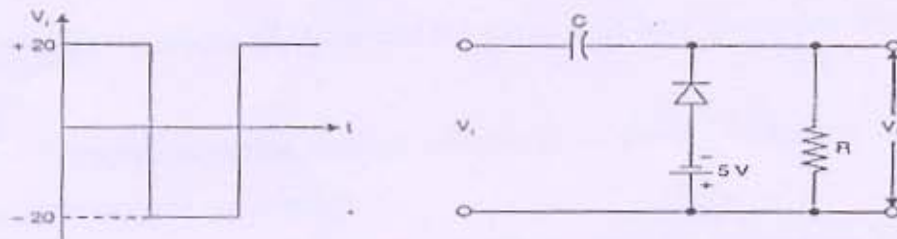


Figure 5
